8 (a) Define electric field.

.....

.....

..... [2]

(b) Fig. 8.1 shows two parallel conducting plates that are in a vacuum. The plates are separated by a distance of 6.7 cm and have a potential difference (p.d.) of 430 V between them.

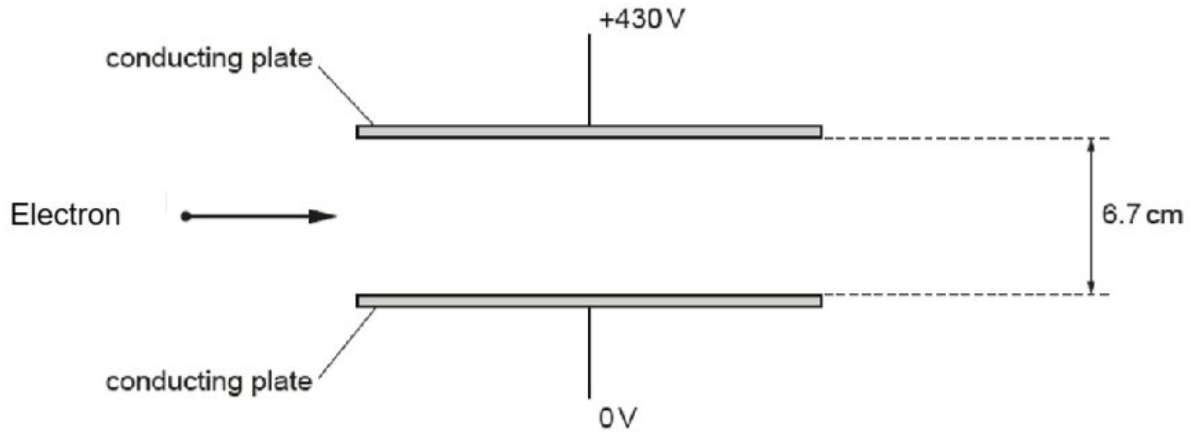


Fig. 8.1

(i) On Fig. 8.1, draw four field lines to represent the electric field between the plates. [2]

(ii) Determine the strength E of the electric field between the plates.

E = ... N C^{-1} [2]

(iii) An electron travels at a speed of 2.6×10^7 m s^{-1} towards the region between the plates, as shown in Fig. 8.1.

On Fig. 8.1, draw the path of the electron as it moves between and beyond the plates. [2]

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(c) A uniform magnetic field is now applied in the region of the electric field in Fig. 8.1, so that the electron in (b)(iii) travels undeviated through the region.

(i) Determine the direction of the uniform magnetic field.

.....

[1]

(ii) Explain, with reference to the forces exerted by the two fields on the electron, why the path of the electron is not deviated.

.....

.....

.....

[1]

(iii) Determine the flux density *B* of the uniform magnetic field.

B = ... T [2]

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(d) State Faraday's law of electromagnetic induction.

.....

.....

.....

[1]

(e) The diameter of the cross-section of a long solenoid is 3.2 cm, as shown in Fig. 8.2.

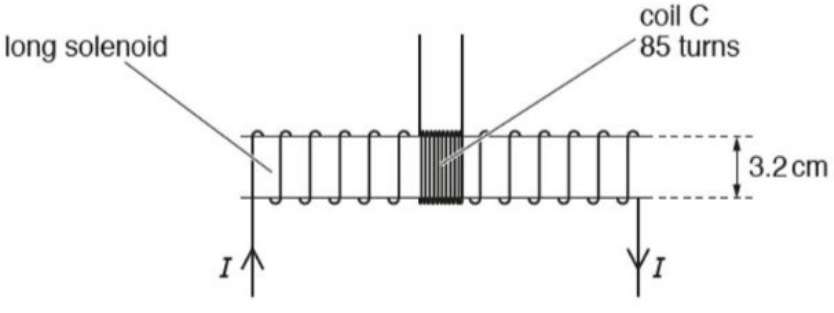


Fig 8.2

A coil C, with 85 turns of wire, is wound tightly around the centre region of the solenoid.

The magnetic flux density B, in tesla, at the centre of the solenoid is given by the expression

$$B = \pi \times 10^{-3} \times I$$

where I is the current in the solenoid in ampere.

Show that, for a current I of 2.8 A in the solenoid, the magnetic flux linkage of coil C is 6.0×10^{-4} Wb.

[1]

(f) The current I in the solenoid in (e) is reversed in 0.30 s.

Calculate the mean e.m.f. induced in coil C.

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e.m.f = ... mV [2]

(g) The current I in the solenoid in (e) is now varied with time t as shown in Fig. 8.3.

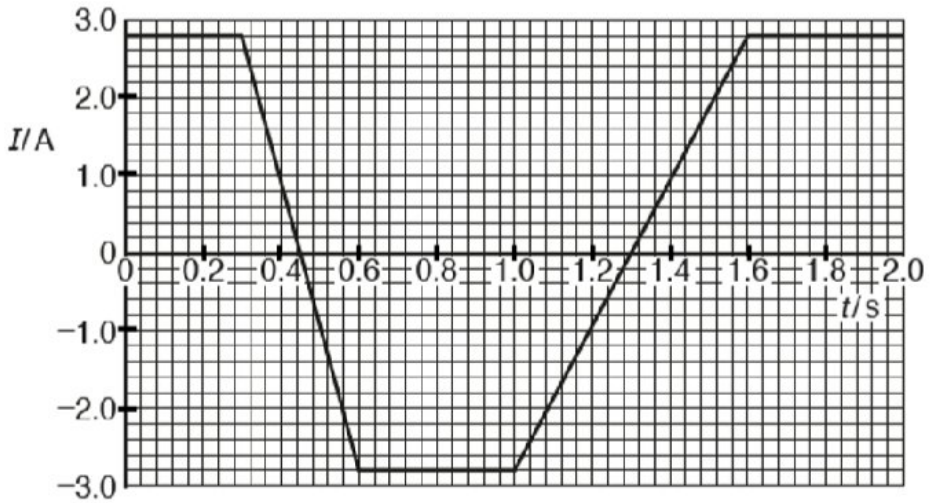


Fig. 8.3

Use your answer to (f) to show, on Fig. 8.4, the variation with time t of the e.m.f. E induced in coil C.

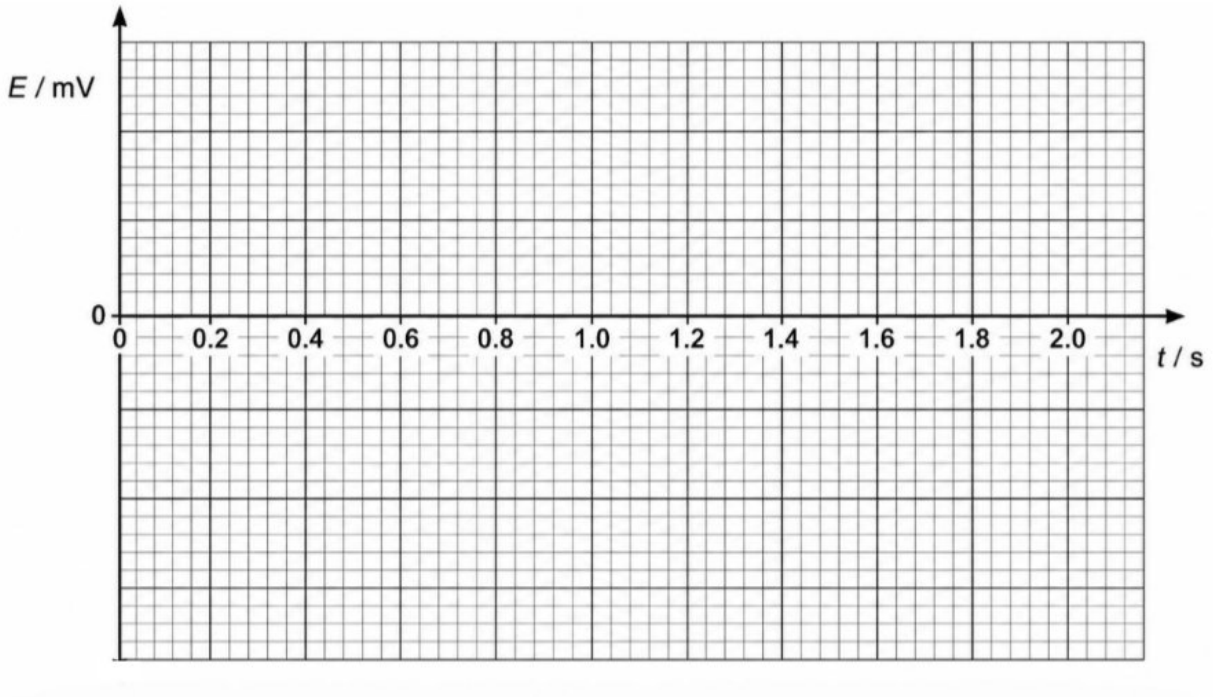


Fig 8.4

[4]

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